

Programme	B. Sc. Mathematics Honours			
Course Code	MAT1VN101			
Course Title	PYTHON PROGRAMMING			
Type of Course	Vocational Minor – Introduction to AI			
Semester	I			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Basic Logic			
Course Summary	Course aims to provide basic programming skills in Python and Python libraries like NumPy etc.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools
CO1	Understand the basics of Python Data structures and Programming constructs	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Understand the basics of Python Programming constructs	U	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Apply Python Libraries for Data Science and Machine Learning	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+ 30)	Ext. Marks (70)
1		Data Types and Data Structures	10	Min.15
	1	Introduction to Python: - using the Python interpreter, Overview of programming in Python		
	2	Expressions and Variables-String Operations.		
	3	Python Data Structures: lists & Tuple –Sets - Dictionaries		
	4	Programming Fundamentals: Conditions and Branching- Loops		
	5	Functions: formal arguments, variable-length arguments		
II		Classes, files and modules	12	Min.15
	6	Introduction to Classes and Objects: -classes, class attributes, instances, instance attributes		
	7	Binding and method invocation, inheritance, polymorphism,		
	8	Built-in functions for classes and instances.		
	9	Files and input/output, reading and writing files		
	10	Methods of file objects, using standard library functions		
	11	Exception Handling		
III		Introduction to Data Science using Python	12	Min.15
	12	Python libraries: Numpy- Scikit- Pandas.		
	13	Importing Datasets: Importing and Exporting Data in Python, Basic Insights from Datasets		
	14	Data cleansing and pre-processing: Identify and Handle Missing Values		
	15	Descriptive Statistics		
	16	ANOVA Correlation		

	17	Dealing with Outliers		
IV		Data Visualization Packages - Matplotlib and Seaborn	11	Min.15
	18	Overview of data visualization concepts		
	19	Introduction to Matplotlib and Seaborn		
	20	Basic Plotting and Customization with Matplotlib		
	21	Basic Plotting and Statistical Visualization with Seaborn		
	22	Other Visualization Libraries – Case Studies		
		Practical's	30	
	1	a) Write a program to calculate compound interest when principal, rate and number of periods are given b) Read name, address, email and phone number of a person through keyboard and print the details		
	2	Write a program to check whether the given input is digit or lowercase character or uppercase character or a special character (use 'if-else-if' ladder)		
	3	a) Print the below triangle using for loop. 5 4 4 3 3 3 2 2 2 2 1 1 1 1 1 b) Python Program to Print the Fibonacci sequence using while loop		
	4	Python program to print all prime numbers in a given interval (use break)		
	5	Write a function called GCD that takes parameters a and b and returns their greatest common divisor		

	6	Write a function called palindrome that takes a string argument and returns True if it is a palindrome and False otherwise. Remember that you can use the built-in function len to check the length of a string		
	7	Define a new class called Circle with appropriate attributes and instantiate a few Circle objects. Write a function called draw_circle that draws circles on the canvas		
	8	Write a python program that defines a matrix and prints		
	9	Write a python program to perform addition of two square matrices		
	10	Python program to perform read and write operations on a file.		
	11	Use the structure of exception handling all general-purpose exceptions		
	12	Write a Python program that calculates basic statistics measures using NumPy		
	13	Create a CSV file named sales_data.csv, which contains sales data for a company. The file has the following columns: Date, Product, Units Sold, and Revenue. Write a Python program using Pandas to perform the following tasks: - Read the data from the CSV file into a DataFrame. - Calculate the total revenue generated by each product. - Determine the total units sold for each product. - Find the date with the highest revenue. - Plot a bar chart showing the total revenue generated by each product.		

14	Create a CSV file named student_grades.csv, which contains the grades of students in different subjects. The file has the following columns: Student_ID, Maths, Science, English, and History. Write a Python program using Matplotlib to perform the following tasks: - Read the data from the CSV file into a DataFrame. - Calculate the average score for each subject. - Plot a bar chart showing the average scores for each subject. - Plot a histogram showing the distribution of scores in Maths.		
15	Visualizing Titanic Dataset You are given a dataset containing information about passengers on the Titanic, including their survival status, age, sex, class, and fare. Write a Python program using Seaborn to perform the following tasks: - Load the Titanic dataset into a DataFrame. - Plot a count plot to visualize the number of passengers in each class. - Plot a bar plot to visualize the survival rate of passengers based on their class and sex. - Plot a heatmap to visualize the correlation matrix of numerical features (e.g., age, fare, and survival status).		

References:

1. Core Python Programming by Wesley J. Chun, 2nd Edition , Pearson Education.
2. An Introduction to Python by Guido Van Russom, Fred L.Drake, Network Theory Limited.
3. Python for Data Science, Dr. Mohd. Abdul Hameed, Wiley Publications - 1st Ed. 2021
4. Python Programming: A Modern Approach, Vamsi Kurama, Pearson
5. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython ,2nd edition, Wes McKinney, O'Reilly Media (2017)

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	1	3	3	2	1	2
CO 2	2	1	3	1	3	3	2	1	2
CO 3	3	2	3	2	3	3	3	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓